YEAR PROJECT PROPOSAL · EMBEDDED SYSTEMS & IOT

Inverter Autopilot

A low-cost smart controller that makes home solar inverters decide for themselves: when to use the battery, when to use the grid, and when to charge.

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Working prototype exists

WHY

In our region the grid supplies only a few hours of electricity a day. Almost every home and small business now depends on a solar inverter with a battery bank to bridge the gaps.

These inverters are not intelligent. They follow a few fixed settings and know nothing about the time, the weather, the house's usage or the battery left for the night. Owners switch modes by hand, batteries run empty before sunrise, grid hours go unused and batteries wear out early.

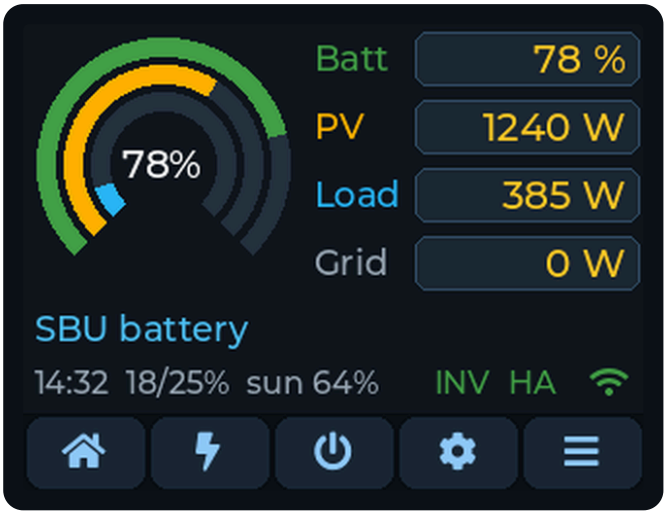
WHAT

A small box with a colour touchscreen that connects to the inverter's communication port. It reads everything the inverter knows (battery level, solar power, house load, grid voltage) every few seconds and takes the decisions an attentive owner would take, around the clock:

Read	Think	Act
Battery, solar, load and grid, plus time and the solar forecast.	Plan the battery reserve and predict the level at sunrise.	Switch battery and grid, charge when needed, sound alarms.

WHAT IT DOES AUTOMATICALLY

- Keeps a larger battery reserve before a cloudy day and a smaller one before a sunny day.
- Moves to the grid when the battery runs low, and back when it has recovered.
- Sends heavy loads (heaters, kettles, pumps) to the grid when the battery cannot carry them.
- Charges from the grid on cloudy days or in an emergency, then returns to solar-only charging.
- Warns with sound and light when the grid is lost or the battery is about to run out.
- Acts only on fresh data, and has a watch-only mode for building trust before taking control.



Home screen, example values. Rings: **battery 78 %**, **solar 1240 W**, **house load 385 W**. Bottom line: time, today's reserve 18/25 %, battery expected at sunrise 64 %, link status.

BUILT FROM

A Wi-Fi microcontroller (Bluetooth LE available), a 2.8″ colour touchscreen with a modern graphics library (LVGL), a serial link to the inverter, a speaker, a status light and a light sensor.

BENEFITS

Low cost
A fraction of the price of a commercial energy manager.

Longer battery life
Fewer deep discharges, sensible charging.

Fewer dark hours
Plans the night instead of reacting to it.

More from the sun
Uses solar first, grid only when it is worth it.

Works offline
All decisions are made on the device itself.

IoT-ready
Connects to phones and smart-home systems when wanted.

How it sees ahead

- It recalculates the battery level expected at sunrise every few seconds, from the charge left, the house's typical load and the time until the sun is useful (about 1.5 h after sunrise if sunny, 3 h if cloudy).
- "Typical load" is the median of the last 3 hours, so a kettle or an hour of heating does not distort the night plan.
- Today's and tomorrow's solar energy come from an internet forecast for these exact panels (size, tilt, direction). Without internet it assumes tomorrow will be like today, and an unknown day counts as cloudy.
- 16 simulated scenarios (cloudy days, outages, false readings, broken links): `160/160` checks passed. In one simulated night it used grid power 17:50–03:30 and kept a third of the battery for the morning.

Talking to the inverter over RS232

- RS232 is a serial standard from 1960, still common in industry because it is simple and reliable over cables up to 15 m.
- The link runs at `2400` bits per second. One short text command returns about 20 values (voltages, power, battery %, temperatures) in a single line of roughly 110 characters, which takes almost half a second to arrive.
- Every message ends with a 16-bit checksum (CRC) to detect corrupted data. If no valid data arrives for 30 s, the controller stops acting.
- RS232 signals swing up to about ±12 V while the microcontroller works at 3.3 V, so a small level-converter chip sits between them. The same protocol is shared by many inverter brands sold in the region.

Next: load automation

Relays that switch water heaters, pumps and air conditioners on solar surplus and shed them when the battery is low.

Then: remote control

See and control the house from a phone, with notifications and daily energy reports.

Later: industrial use

Clinics, shops, farms, telecom sites and water pumping: many sites managed and compared from one place.